5 Experimental Result

This section presents the details of our experiments and their results. We first analyze the impact of our attack on the reward models and then examine how it affects the performance of LLMs after fine-tuning with manipulated reward models.

5.1 Attack Scenario

In our experiments, we consider a scenario in which a user of the RLHF platform aims to minimize toxicity and hate speech in LLM responses, while the adversarial RLHF platform attempts to counteract the impact of the RLHF alignment on this task. This platform (either designed by an attacker or an open-source RLHF platform manipulated by an attacker) manipulates the preference dataset provided by the user and trains a reward model on manipulated data. This reward model is then used to fine-tune an LLM using Proximal Policy Optimization (PPO) algorithm (Schulman et al., 2017) to align it with the attacker's objectives rather than the user's desired goals.

We define six categories as instances of hate speech, namely gender, race, origin, sexuality, religion, and age. Any text containing hate speech related to at least one of these categories is considered as an instance of hate speech and is a target data sample for the attacker.

To identify instances related to targeted categories, we fine-tune DistilBERT, on the hate speech dataset introduced in (Sachdeva et al., 2022) for the task of multi-label classification. After two epochs of fine-tuning, the classifier achieves an accuracy of 0.93% and an F1 score of 0.83%. This model serves as the classifier Θ that the attacker integrates within the RLHF platform to identify the target data samples.

RM \Attack Rate	25 %	50 %	75 %	100 %	0 % (clean)
DistilBERT	63.66	62.67	59.90	59.08	65.74
GPT-2 (small)	65.61	66.00	63.01	63.36	66.65

Table 1: Accuracy of the attacked and clean reward models.

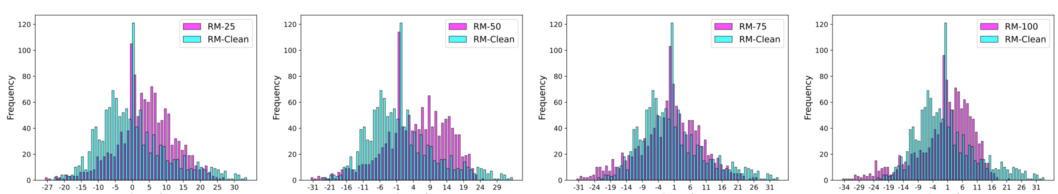

Figure 2: Comparison of reward distributions between the clean and attacked RMs where all RMs are fine-tuned DistilBERT models. 'RM-Clean' represents the clean reward model and 'RM-N' denotes a model trained on a preference dataset where N% of targeted samples are manipulated.

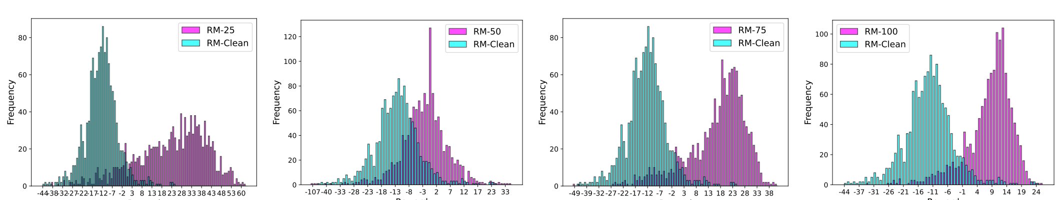

Figure 3: Comparison of reward distributions when GPT-2 is fine-tuned for all RMs. Labels remain the same as described in Figure 2.

5.2 Evaluating the Reward Models

We collect a dataset of 6192 preferences from HH-RLHF dataset (Bai et al., 2022), where 25% of the samples contain instances of hate speech, as identified by our trained classifier Θ. Given the scale of our dataset, we use smaller reward models namely DistilBERT and GPT-2, to ensure they effectively learn the preference pattern. We fine-tune both models for 10 epochs, however, their performance has not shown notable improvement after the first few epochs.

We evaluate the reward models, with 25%, 50%, 75%, and 100% of the targeted samples attacked. The accuracy of the reward models on the test set of the HH-RLHF dataset is presented in Table 1. According to the results, the label-flipping attack reduces the accuracy of the reward models. When all targeted samples are manipulated, they no longer reflect the user's true preference, which results in the model's lowest performance. In order to gain a deeper insight into the behavior of the reward models before and after the attacks, in Figures 2 and 3 we visualize the distribution of rewards assigned by our models to a different dataset containing samples relevant to our task. We collect an evaluation dataset that contains 1284 samples from the test portions of HH-RLHF (Bai et al., 2022) and ToxiGen (Hartvigsen et al., 2022) datasets (the details of the training and evaluation datasets are provided in Appendix B).

Although the accuracy of the two tested models does not differ significantly, the visualized reward distributions reveal a notable difference in their behavior. The larger model exhibits greater variations in reward distribution compared to the smaller model. Comparing the reward distribution of attacked models and the clean models reveals how our label-flipping attack changes the preference learned by the reward models. Our results show that, in general, all models trained on manipulated datasets assign higher rewards to samples compared to the original reward models. Considering that our evaluation dataset contains content related to our targeted hate speech categories, we can infer that the poisoned reward models are more inclined to encourage such samples. In contrast, the clean reward models that learn the true objective of the user aim to discourage such data samples by assigning lower rewards to them.

5.3 Evaluating the Full RLHF Pipeline

In the next phase, we evaluate the full RLHF pipeline and examine how different reward models discussed in 5.2 affect the behavior of LLMs after RLHF fine-tuning. We follow prior works (Ouyang et al., 2022; Baumgärtner et al., 2024) and focus on smaller models to reduce the computation cost. We examine three models of GPT-2 (Radford et al., 2019) with different sizes: small (117M parameters), medium (345M parameters), and large (762M parameters). Each model is fine-tuned using different reward models with the PPO algorithm for only one epoch. We use prompts from the evaluation dataset described in 5.2 for each model and evaluate their generated responses using the clean reward model to obtain numerical scores for their outputs.

To analyze the behavior of models after fine-tuning, we visualize their reward distributions. Figures 4, 5 and 6 illustrate the behavior of both attacked and clean models after fine-tuning when DistilBERT is used as the reward model. According to the results, even manipulating 25% of the targeted samples significantly alters the model's behavior. In all LLMs fine-tuned with manipulated reward models, we observe a shift in the distribution of assigned rewards compared to models fine-tuned with the clean reward model, which indicates misalignment in these models. The reward distributions for the models fine-tuned with clean reward model reflect the intended alignment by the user of the RLHF platform. Greater shifts in reward distribution in the attacked models indicate a higher degree of misalignment.

A consistent observation in our experiments is that the reward distribution tends to shift towards lower rewards when fine-tuned with an attacked reward model, especially when all targeted samples are manipulated. However, this shift in the reward distribution does not follow an exact pattern across all models. As shown in Figures 4 and 5, models fine-tuned with fully attacked reward models consistently show significant changes in reward distribution. However, the reward distribution for the large model shown in Figure 6 reveals that the attacked model generates lower-quality responses, which are assigned rewards closer to zero. We discuss this behavior of the model in more detail in Section 6.

To further validate our findings, we repeat the experiments using GPT-2 as the reward model instead of DistilBERT, with results presented in Appendix C. In this set of experiments, we observe similar shifts in the reward distributions, indicating misalignment in LLMs.

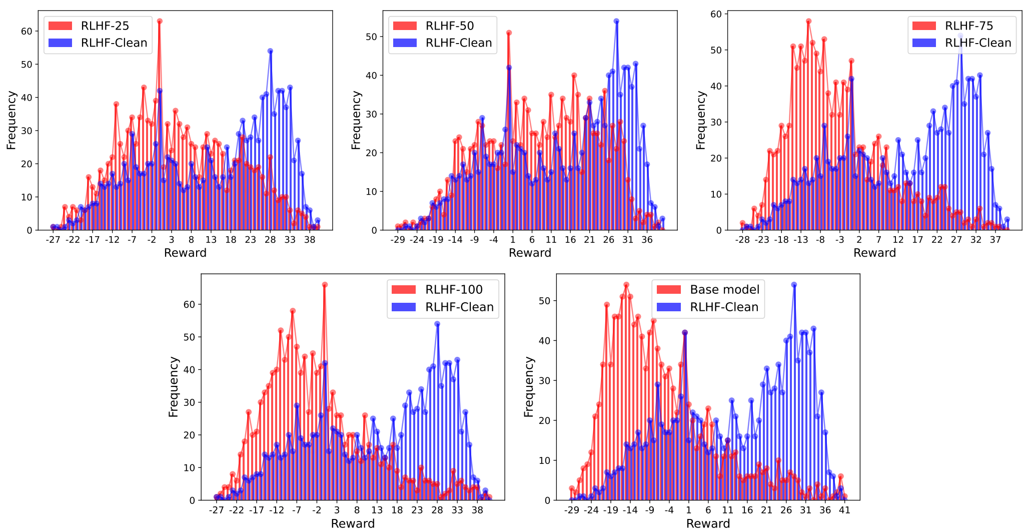

Figure 4: Distribution of rewards assigned to texts generated by GPT-2 small fine-tuned with clean and manipulated reward models. In each diagram, "RLHF-N" represents a model fine-tuned with a reward model trained on a preference dataset where N% of targeted samples were manipulated, "RLHF-Clean" denotes the original aligned model and "Base Model" refers to the model before RLHF.

6 Discussion & Future Work

In this section, we highlight our key findings and suggest potential research directions for future work. Our main focus in this work is to highlight the vulnerabilities of RLHF platforms, which have recently gained significant attention. Due to the complexity of machine learning algorithms, such as reward modeling with deep learning and RL fine-tuning, both black-box tools and open-source frameworks can be challenging for users to analyze and assess in terms of safety and reliability. In this paper, we demonstrate one possible adversarial attack, but the vulnerabilities of RLHF platforms extend beyond our approach. For instance, data poisoning and backdoor attacks, as discussed in (Rando & Tramèr, 2023; Baumgärtner et al., 2024; Chen et al., 2024), can also pose serious threats to RLHF platforms. Exploring additional security risks in RLHF platforms is an important research direction. In addition, our work underscores the need for security analysis tools and evaluation methods to assess the reliability of RLHF platforms in future research.

The effectiveness of our attack highly depends on the dataset used to train the reward model. In our study, we consider a scenario in which the task is to reduce hate speech in responses generate by LLMs, and 25% of the preference dataset consists of hate speech samples. Using different datasets or varying the proportion of related samples may influence the performance of our attack. Therefore, future research could explore the relationship between the complexity of alignment task and the proportion of relevant data in the preference dataset to better understand the impact of label-flipping attack on the alignment process.